

SELF-SERVICE REPORTING WITH SLICERDICER

Quick Start Guide

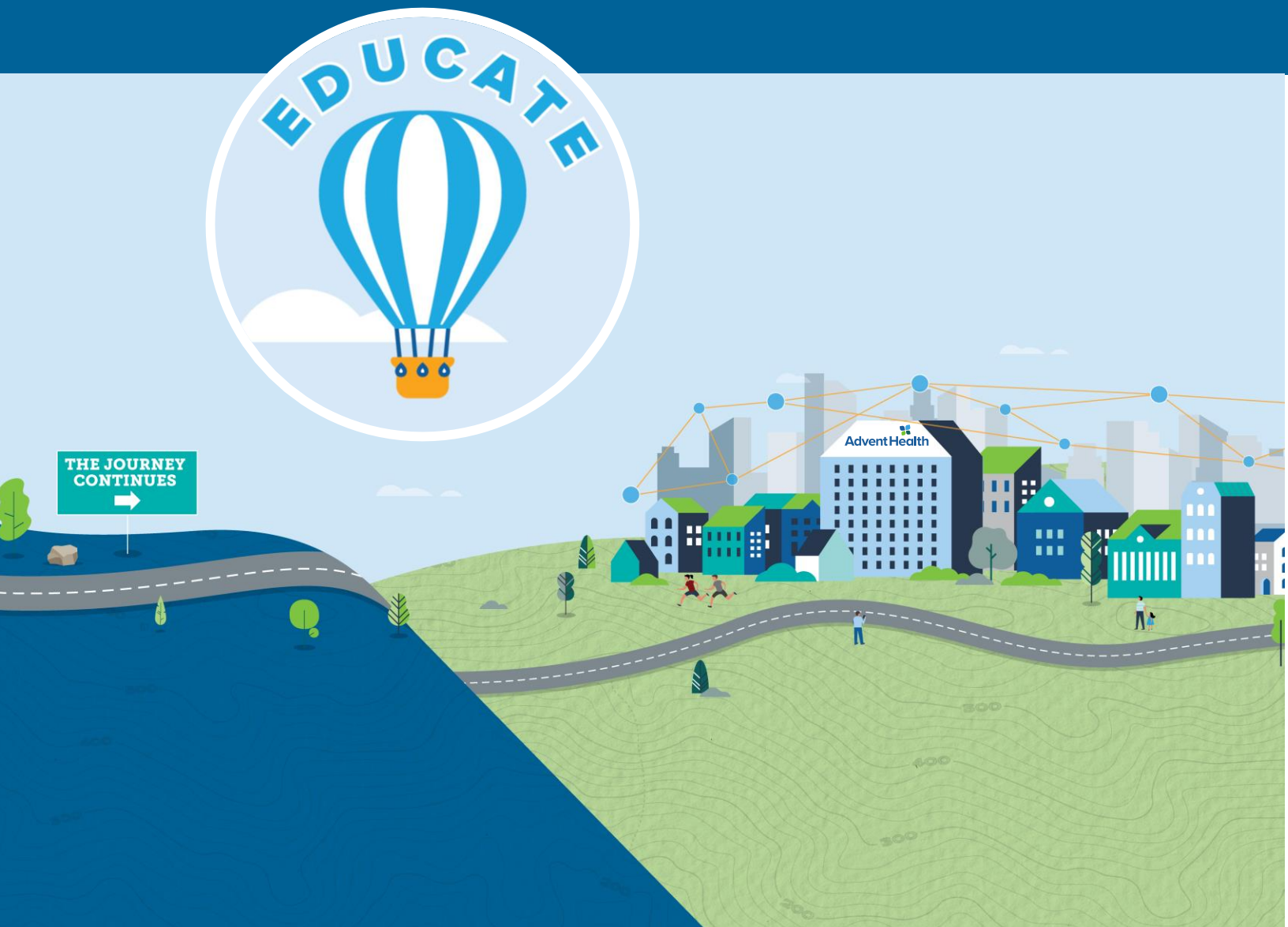




TABLE OF CONTENTS

SELF-SERVICE REPORTING WITH SLICERDICER3

 ACCESS SLICERDICER3

 GETTING STARTED WITH THE ESSENTIALS TUTORIAL3

DEFINE A POPULATION3

 CHANGE THE ACTIVE DATE RANGE5

 CREATE A COMBINATION FILTER6

 ADD ANCHOR EVENT CONFIGURATION6

 CHANGE THE LOGIC BETWEEN FILTERS7

 CREATE A POPULATION BASED ON MULTIPLE DATA MODELS9

 CONFIGURE ADVANCED OPTIONS ON YOUR FILTERS9

 Add Sequential or Overlapping Criteria9

 Add multiple values11

 Specify the Age at Time of Event11

 MODIFY EVENT LINKING12

 FOCUS ON RECENT DATA13

 EXPORT PATIENTS TO REPORTING WORKBENCH13

SAVE AND SHARE SESSIONS14

 SEE AND EDIT SAVED SESSIONS14

 SHARE A SESSION14

SLICE A POPULATION TO MAKE COMPARISONS14

 CREATE CUSTOM SLICES15

 GRAB ONLY CERTAIN SLICES15

ADD A MEASURE15

 ADD GOALS AND THRESHOLDS TO YOUR MEASURES18

 ANALYZE THE VARIABILITY OF YOUR DATA WITH CONFIDENCE INTERVALS18

 CREATE CUSTOM MEASURES20

CHANGE VISUALIZATION OPTIONS20

 PICK THE RIGHT VISUALIZATION FOR YOUR ANALYSIS20

 VIEW DETAILS ABOUT RECORDS IN THE DETAIL TABLE21

 SEE RELATIONSHIPS WITH THE CROSTAB VISUALIZATION22

 SEE POPULATIONS ON A MAP22

 MODIFY THE GRAPH'S AXIS DEFINITIONS22

 CUSTOMIZE YOUR GRAPH COLORS23

HIDE NONE OF THE ABOVE, SAVE CLICKS, AND SHARE DATA24

SLICERDICER SHORTCUTS25

 ALWAYS APPLICABLE25

 TEXT EDITING25

 USER INTERFACE26

 DETAIL TABLE SHORTCUTS26



DISCOVER SLICERDICER USING EXPLORER DASHBOARDS27

TRY IT OUT 27

ADDITIONAL RESOURCES..... 28





SELF-SERVICE REPORTING WITH SLICERDICER

Use SlicerDicer to find data you need to investigate a hunch. You can refine your searches on the fly to better understand the data. Examine trends, drill down to line-level details, and jump to related records to follow up directly in Hyperspace.

Access SlicerDicer

Launch the Analytics Catalog and search for SlicerDicer content, including data models and sessions.

- You can launch that content in SlicerDicer directly from the catalog. If you're unsure what to select, you

can hover over content and click  to view more information about it.

- If you cannot find your content, try applying the SlicerDicer Content Type filter.

Alternatively, use the Epic assistant bar to search for SlicerDicer.

- Select a data model.
- If you're unsure which model to select, hover over the data model and click  for more details.
- If you only have access to one data model, that model opens automatically.
- If you select the wrong data model, click **Start Over**  in the toolbar to clear the graph and select a new data model.



SLICERDICER IS A GREAT TOOL BUT IS NOT THE BEST CHOICE FOR ALL REPORTING NEEDS. SLICERDICER DRAWS FROM DATA THAT IS AT LEAST A DAY OLD, SO YOU CAN'T USE IT FOR REAL-TIME REPORTING, SUCH AS ACUTE CARE MONITORING. IN ADDITION, MOST OF THE TIME SLICERDICER CAN SEARCH ONLY DISCRETE DATA. FOR EXAMPLE, SLICERDICER DOESN'T KNOW A PATIENT HAS A PARTICULAR DIAGNOSIS IF THE ONLY PLACE THE DIAGNOSIS IS DOCUMENTED IS IN A FREE TEXT NOTE.

AS ALWAYS, HANDLE PATIENT DATA WITH CARE.

Getting started with the Essentials tutorial

SlicerDicer has several Essentials tutorials to help you discover the core features, which are accessible in every data model using the Tutorials menu in the upper right corner of the screen. Many data models also have customized Story tutorials that walk you through common workflows specific to the data model.

1. Choose a data model from the Analytics Catalog or the SlicerDicer data model selector.
2. In the upper right corner of the screen, click  **Tutorials**.
 - Choose a data model-specific tutorial from the Story Tutorials list for a more in-depth walkthrough.
 - Select the SlicerDicer Essentials #1 tutorial for a walkthrough of the basic features of SlicerDicer.

DEFINE A POPULATION

The population you can see by default in any given data model depends on your user security and the base filters configured for that data model. All users can view results for any base filters on the model, like My

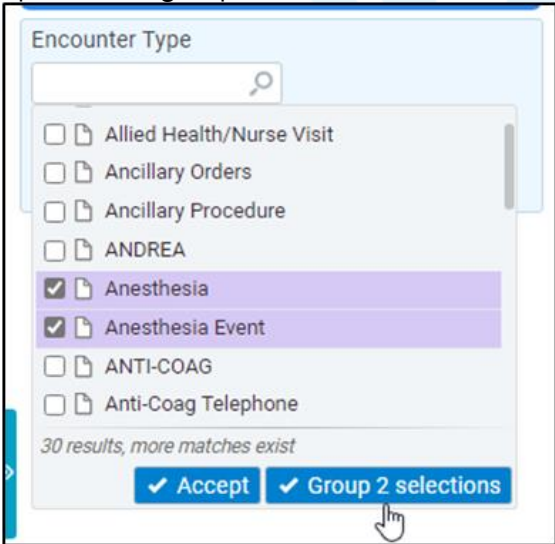


Patients, My Departments, or My Providers, but some users can view All Records. If a given data model has Base filters available, they appear in the Population section:



Use criteria to further narrow your starting population, filtering on items such as diagnoses, weight, or lab components. You can add multiple criteria, add sequential or overlapping criteria, and use additional advanced logic on criteria to get to the data you need.

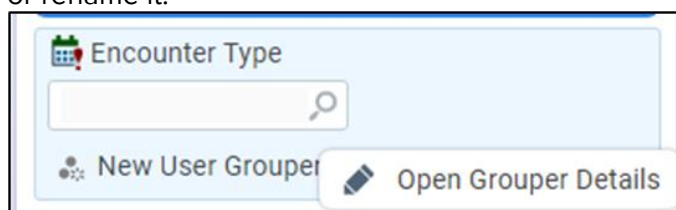
1. Launch SlicerDicer and select a data model (for example, Patients).
2. Click **Browse** to search available criteria or use the search bar.
 - The search bar in the population section can find both filters and options (for example, individual diagnoses rather than simply the Diagnosis filter).
3. Select a criterion. If you want to learn more about a criterion, click [i](#).
4. After selecting a criterion, you may need to provide additional details:
 - For criteria based on lists of records or category values, like Diagnosis, you need to select at least one option, like Hypertension. If you don't select an option, the criteria isn't active and doesn't affect your results.
 - You can select multiple options at once using Shift+click.
 - If you want to select many records that share a similar name, search for a common term, select the options and group them:



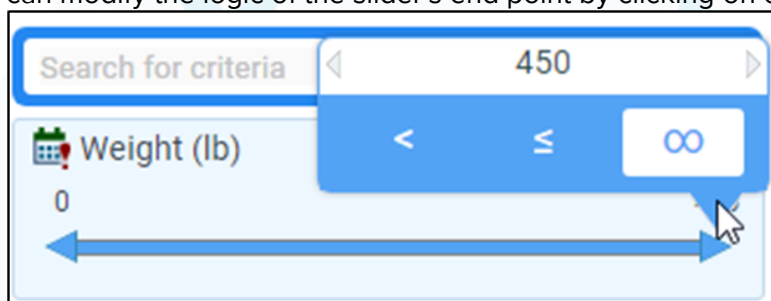
- Once grouped, you can edit, rename, or save your custom grouper as needed.



- Custom groupers are not automatically updated, so if new values might match the search terms you used to create the grouper, you should manually update the grouper by creating a new grouper based on the original grouper including the new values.
- Right-click the grouper and select **Open Grouper Details** to remove values from the grouper or rename it.



- For numeric criteria, like patient Weight, you need to choose a range on a slider bar. By default, SlicerDicer shows patients with any value documented, even beyond the range on the slider bar. You can modify the logic of the slider's end point by clicking on one of the values.



- You can move the stops on the slider bar to further narrow the range.
- Most numeric slider bars split the range evenly, but some are configured to use more meaningful ranges. For example, the BMI filter uses NIH Standard as its default slice range. When multiple ranges are available, you can select your desired range from the available list. You can always customize the range.
- For Boolean criteria, like Pregnant?, SlicerDicer automatically finds records that match the criteria. All Boolean criteria return two values - true and false.
- For free text criteria, like Component Result Text in the Lab Specimens and Tests data model, you need to enter a keyword or phrase for SlicerDicer to search for.

5. To exclude records that match the selected criterion, click and select **Exclude**.



Remove a criterion by hovering and clicking X.



Click Undo (Ctrl + Z), Redo (Ctrl + Y), or Start Over in the toolbar to fix any mistakes.

Change the active date range

Some data models have more than one relevant date for the query date range. For example, the ED Encounters data model can show results based on the ED encounter date, arrival date, or departure date. To change the date range used for your session:

1. Click the **Dates** section.
2. Select a new date range by clicking in the **Based On** field. Not all data models have multiple date ranges, so you might not see options here.



3. If your session has dates, you can divide the results by dates as well. By default, no date slices are applied. To add a date slice, choose an option in the **Slice By** field. You can choose from year, quarter, month, week, day, or none.

Some data models, like Patients, do not have associated date ranges. In these data models, dates are only added to your session if you add at least one filter with its own dates.

 **Dates**

Range: All Time 

For example, adding a slice by Admitted? adds dates to your Patients session since the admissions themselves have associated dates. Multiple filters with different date ranges can be active at the same time. The results for each filter are compared to their own date ranges.

Create a Combination Filter

Combination filters allow you to save a set of population criteria into a single concept. You can then reuse that combined filter logic for multiple reports, reducing the redundancy of creating multiple sessions with the same set of criteria. To create a combination filter:

1. Select a data model which does not have date ranges, such as Patients.
2. Select at least two population criteria you want to include in your combination filter.
3. Select your filter values if applicable.
4. Click  in the Population section header.
5. Click  **Select Criteria** to create a combination filter with existing criteria in your session.
6. With your criteria selected, click  **Create Combination Filter**.
7. Enter a name for the filter.
8. Click **Save**.

Add Anchor Event configuration

Anchor events give you another way to investigate how events relate to each other. An anchor event is a specific occurrence you define as a fixed reference point for the results you're interested in investigating.

1. Click the **Dates** section and change the date type from Calendar to **Relative to Event**.
2. In the **Anchor Event** field, select the filter or date range you want to use.
3. Use the **Start Date** and **End Date** fields to define the time range the anchor event occurs within.
4. In the **Based On** field under Relative Time Range, select the filter or date range you want to compare to the anchor event.
5. Use the **From** and **To** fields to set how long before or after the anchor event the based on event occurred.
6. Use the **Slice By** field and dropdown menu to set how much time each slice represents.



1 **Dates**

Date Type

2 Anchor Event Time Range ⓘ

Anchor Event
Visit Date

3 Start Date
4/12/2023

End Date
10/11/2023

4 Relative Time Range ⓘ

Based On
Appointment Creation Date

5 From
0 Months

To
6 Months

6 Slice By 1 Month

☐ Slice Offset ⓘ

Change the logic between filters

SlicerDicer allows both simple and advanced logic between criteria in the population card. You can toggle between AND and OR logic between your filters by clicking on the logic separator:

Family History

► Cancer

Last Stored Data

Admitted? **AND**

Department

You can also toggle the logic between options within criteria based on lists of records:



Family History

▸ Cancer

▸ Diabetes insipidus

▸ Diabetes mellitus

OR

OR

Last Stored Data

SlicerDicer also offers an advanced logic mode for more complex scenarios. You might want to do this if you want to return a very specific set of results that cannot be replicated with AND or OR logic between each filter. To enable advanced logic mode, click the toggle at the bottom of the Population section in the sidebar:

Advanced Logic

In Advanced Logic mode, each filter shows an associated number you can use to define your logic. This is useful for grouping filters into sets, like in this example:

2 📅 Lung Screening Patient?

1 📅 Diagnosis

▸ 🍷 Lung cancer screening decl...

3 📅 Medical History

🍷 Smoking/drinking/substanc...

Last Stored Data

Logic: 2 OR (1 AND 3)

Default Logic



Create a population based on multiple data models

The answers to some questions require data from multiple data models. For example, you might want to review patient data associated with transactions. If this data is not available in your current data model, you can link to additional models to investigate further.

1. Define your population in the starting data model.
2. Click  in the **Population** section.
3. Choose the data model where you want to continue your investigation.

Configure Advanced Options on your filters

The following topics allow you to further filter to a subset of your population.

Add Sequential or Overlapping Criteria

Sequential or overlapping comparisons between events can be done in any data model which is related to patients. As an example, the Referrals data model supports sequential comparisons because we can compare referrals for the same patient, but the Logistics Jobs data model does not support them, because logistics jobs are not directly related to patients.

These features compare filter event dates in data models without a date range, like Patients, Patients with Cancer, Breast Imaging Lesions, or Pharmacogenomic Diplotypes. In data models with date ranges, like Encounters, the comparisons occur between separate data model events, like different encounters, admissions, or medication administrations.

Sequential Criterion

Suppose you're investigating root causes of opioid addictions and wondering whether having surgery within the last year could be a factor. SlicerDicer can show you the patients who had a surgery and were subsequently diagnosed with opioid dependence.

1. Add a criterion for the event that occurred first. In this example, filter results by an encounter type of surgery.
2. Right-click the criterion and select  **Add Sequential Criterion > Add New Criteria**.
3. Select the criterion for the event that occurred second.
 - a. In this example, select the **Diagnosis** criterion and enter opioid dependence.
4. Update the **Within** field for your use case.
 - a. In this example, enter 12 months.

Overlapping Criterion

Overlapping date ranges require two criteria to have occurred simultaneously. For example, suppose you want to find pregnant patients who had a specific procedure. For this report, right-click the **Procedures** criterion, select **Add Overlapping Criterion**  **> Add New Criteria**, and select the **Pregnant?** filter. If you don't specify the criteria as overlapping, SlicerDicer returns patients who were pregnant and had the specified procedure at any time during the query date range, even if, for example, the pregnancy didn't begin until after the procedure occurred. If you want to see sequential rates, the workflow for adding the criteria is slightly different than adding sequential criteria in your main population. For example, suppose you want to see the percentage of patients who were discharged and had an admission within 30 days.

1. Add a criterion for the event that occurred first.
 - a. In this example, add the Discharged? criterion.
2. Add a measure for the event you want to see the rate of and verify the Percentage calculation is selected.
 - a. In this example, add the Admitted? measure.
3. In the right pane, click **Make Rate Sequential**.
4. In the First section, choose the event that occurred first.



- a. Choose the Discharged? criterion in this example.
5. In the And Then section, update the Within options to match the time frame for your use case.
 - a. In this example, enter 30 and select Days.

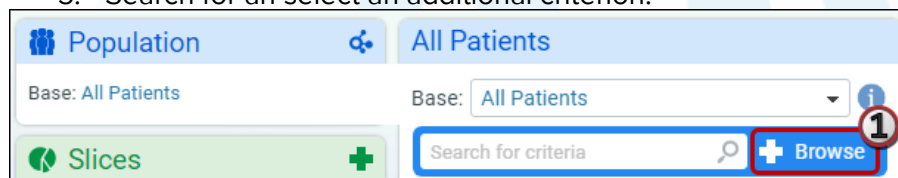
Not all criteria track data over time. If a criterion has a Last Stored banner, that means it uses only the most recent value. This has the following implications for SlicerDicer:

- The query date range doesn't apply to these criteria. For example, suppose your session includes only the MyChart Status criterion. If you try to use the line graph to see how the number of patients with an activated MyChart account has changed over time, the line will always be flat because SlicerDicer uses only patients' current MyChart status.
- Specifying sequential or overlapping criteria doesn't apply to these criteria in data models without date ranges, like Patients. They can be used in sequential comparisons in data models with date ranges.
- Using a different date range than the range on the graph doesn't apply to these criteria.
- Slice values for these criteria don't change across date slices.

Add Sequential or Overlapping Criterion directly

You can also add sequential or overlapping criterion directly from the Add Criterion menu.

1. Click **+ Browse**.
2. Select **Sequential** at the top of the Add Criterion menu.
3. Search for and select a criterion.
4. Click **Add Sequential Criterion** or **Add Overlapping Criterion**.
5. Search for and select an additional criterion.



Control the order of events

- You can also control the order of events after configuring the criteria in the chain. You can choose from Followed By, Preceded By, Not Followed By, or Not Preceded By.
- The Not Followed By and Not Preceded By options can be used only once in the chain and only at the end of the chain.
- Click the banner between any two events to change your selection for the entire chain.



Hospital Encounter

Base: All Patients

Search for criteria

Same encounter

Encounter Type

► Hospital Encounter

Followed By

Switch to Preceded By

Switch to Not Preceded By

Switch to Not Followed By

selection is made.

Add multiple values

You can add multiple values to a SlicerDicer filter at one time when creating reports in SlicerDicer. If you have an external list of values such as procedures, medications, or diagnoses, you can search for and add all of the values at once instead of one at a time. You can also save these values into a user grouper for quick access in the future.

1. Add a population criterion that lists values, like diagnoses or procedures.
2. Hover over the criterion and click  to open the Advanced Options menu.
3. Select  **Add Multiple Values**.
4. Search for a list of values in the **Search Terms** field.
5. Select the delimiter that is used to separate the search values or type a custom delimiter into the text box.
6. If you are searching for specific values, select the **Exact Matching** checkbox.
7. Click **Next**.
8. If you do not want to include a matched value, clear the checkbox to remove that value from your session.
9. Select the **Create User Grouper** checkbox to create a temporary user grouper from the options that you have selected.
 - a. This is beneficial when you are searching for multiple values that can be treated as a group.
10. When you're satisfied with your search terms, click one of the **Accept** or **Accept & Add More**.
 - a. **Accept** closes the window and saves your groupers.
 - b. **Accept & Add More** reopens the first window, allowing you to create additional user groupers.

If you created a temporary user grouper and want to use the grouper you created for other searches and reports, save the grouper.

11. Right-click the grouper and select **Open Grouper Details**.
12. Add a description and click **Save As**.

Specify the Age at Time of Event

In the Patients, Patients with Cancer, and Breast Imaging Lesions data models, you can filter your results by the patient's age when an event was recorded.



In these models, add a filter with dates, like Encounter Type, and select at least one Encounter Type. To narrow results to patients who were a specific age when the encounters occurred:

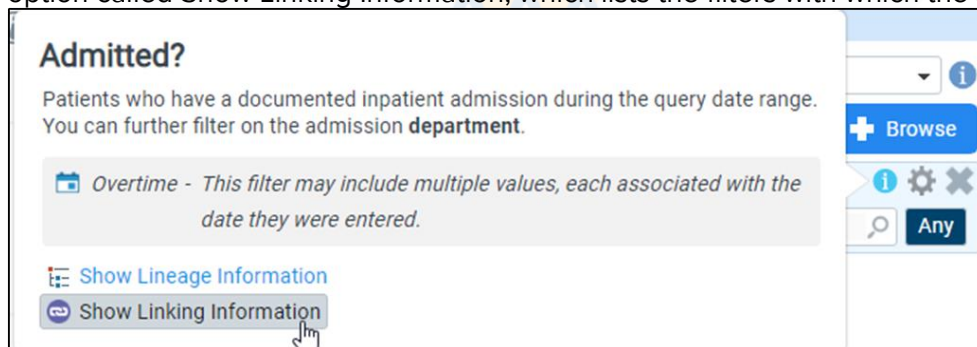
1. Hover over the criterion and click  to open the Advanced Options menu.
2. Select **Specify Age at Time of Event**.
3. Choose the age range relevant to your search. Options include years, months, weeks, and days.
4. You can further modify this behavior with the **Refine Search** options:
 - a. Enabling the first checkbox returns results from all time.
 - b. Enabling the second checkbox is relevant only to graphs with date slices. Select this option to ensure the patient was in the specified age range within each given date slice, rather than at any point in the date range on the graph.

Modify Event Linking

Linking indicates that the results of two or more filters were recorded during the same event, like an encounter. Linking allows users to examine related results more easily. To configure these options, hover over the criterion

and click  to open the Advanced Options menu..

- Filter linking requires two or more criteria to be associated with the same event. For example, suppose you're using the Patients data model and you add the Department criterion for the North Department and the Encounter Type criterion for office visits.
 - If the criteria are linked together, as shown by the , SlicerDicer returns patients who had an office visit in the North Department.
 - If you unlink the criteria, as shown by the , SlicerDicer returns patients who had any type of encounter in the North Department if they also had an office visit in any department.
- Population filters and slices that can link do so by default. Measures that can link do not link by default, but you can choose to link them. Measures cannot link to only measures – you must include at least one population filter or slice.
 - Many filters cannot link to other filters. Linking is more common in data models that have many filters representing events that did not occur on the data model's root event (for example, an Encounter is the root event of the Visits data model, while an Admission or Discharge is a separate event).
 - If a filter cannot link to any other filter in the session, no linking icon appears. After you add at least two linkable filters, the icons appear. There are many data models in which no filters can link.
- To check if a filter can link, you can check the information pane. Filters that can link include a menu option called Show Linking Information, which lists the filters with which the selected filter can link.





Linking also applies to certain columns in the detail table. If a filter can link, it can be linked to other filters in the detail table. This does not change the results, but it does restrict the values you see in the table. Only those column values that occurred in the same event appear when columns are linked. Note that this formatting cannot be exported to Excel.

Same encounter:

Admitted?

Department

Linked by encounter

Admitted? and Department were recorded in the same encounter

2,228

MYC PST DEFAULT DEPT	
MYC PST DEFAULT DEPT	
MYC PST DEFAULT DEPT	
DDW FAMILY MEDICINE	
WI HBN FAMILY PRACTICE	
DDW FAMILY MEDICINE	
JLB FAMILY PRACTICE	
ANQA SURGERY MELBOURNE	
FAMILY PRACTICE NORTH	
DDW FAMILY MEDICINE	
INITIAL DEPARTMENT	
WI HBN FAMILY PRACTICE	
Yes	MONONA RESCARE

Focus on recent data

Suppose you want to find patients who were diagnosed with a condition for the first time in the last year. You can't simply add a diagnosis criterion and change the date range to be the last year, because that population would include any patients who had the diagnosis in the last year, regardless of when they were first diagnosed. To create a report like this one, use a sequential criterion with Not Preceded By logic.

These steps describe this setup in more detail:

1. Open SlicerDicer and select the Patients data model.
2. Add the Diagnosis criterion and select the condition you're interested in.
3. In the Dates section, change the start date to y-1.
4. Right-click the diagnosis criterion and choose **Add Sequential** criterion.
5. Select another diagnosis criterion with the same option you selected in step 2.
6. Click the Followed By or Preceded By banner between the criteria. Change the logic to use **Not Preceded By**.

Export Patients to Reporting Workbench

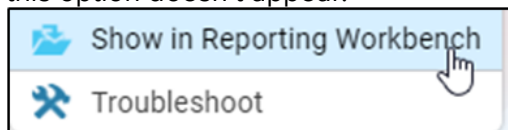
This feature is unique to the Patients data model. You can define a population of patients in SlicerDicer, then export those patients to Reporting Workbench for further exploration. Note that Reporting Workbench might have more recent data than SlicerDicer. This might result in a different number of patients in your Reporting Workbench population than in SlicerDicer.

To export patients to Reporting Workbench:

1. Create or load a population in the Patients data model. Modify your filters or date range as needed.
2. Right click the graph.



3. Select the Show in Reporting Workbench menu option. If you don't have Reporting Workbench security, this option doesn't appear.



4. Your population opens in Reporting Workbench. The columns are populated automatically for all exports. Contact your SlicerDicer administrator if you feel columns are missing.

SAVE AND SHARE SESSIONS

You can save a session if you use it frequently and might want to revisit it later.

1. On the toolbar, click  **Save As** and give the session a name.
2. Review and change additional information as needed, such as the date range, and click **Create Session**. You can add the session to a group to organize your Load screen.

See and edit saved sessions

1. From the Analytics Catalog, search for your session and launch it.
2. Alternatively, within SlicerDicer, click **Load**  on the toolbar and select the desired session. You can create, rename, and reorder groups for your saved sessions on the Load screen or when saving a session. You can have four (4) custom groups. The Shared and Published groups appear for all users.
3. Modify the session as needed and save changes.

Share a session

1. On the toolbar, click  **Share**. Note that sharing automatically saves your session if you had not saved it previously. Sharing an existing session creates a new copy of the session.
2. Enter a name and select how you want to share the session and with whom. If you choose to share with specific people, enter each user you want to give access to the session. Options for sharing include:
 - a. Grant Access. You can share a session with specific users. Search by user name or ID. Some users can also publish sessions (share with everyone).
 - b. Secure Chat. You can send a direct link to the session to specific users over Secure Chat.

SLICE A POPULATION TO MAKE COMPARISONS

Use slices to split up your population to compare results among subgroups. For example, suppose you use criteria to narrow your starting population to patients with an active MyChart account. You could then slice this population by county to see which counties have the most MyChart users.

When you add slices, you can grab the top or bottom slices, such as the 10 counties with the most MyChart users. You can also enter specific slices, such as to compare the number of users in Dane, Jefferson, and Dodge counties.

Here are some other tips for working with slices:

1. Click  next to Slices to add a slice.
2. Select the appropriate filter.
3. Click the magnifying glass to select specific slices. Alternatively, click  to select the top or bottom slices for your current measure.
4. Repeat steps 1-3 as many times as you'd like to divide your population.



5. If you have similar slices you want to group together, select the check boxes next to those slices and click **Group**.
6. You can rename slices by clicking on the slice name. To reset the name back to the original, remove and re-add the slice.
7. You can remove slice options from your session by hovering over the slice and clicking . For numeric options, use the checkbox to the left of the range.
8. You can remove a slice from your session by hovering over the slice and clicking .
9. If you want to temporarily hide a slice on your graph, hover over the slice and click . This hides all slices in the group, not just individual options.
10. You can click and drag slices to reorder them.
11. Whether or not SlicerDicer adds a None of the Above slice to list slices by default is controlled by a user setting as described in the [Hide None of the Above, Save Clicks, and Share Data](#) topic below.
12. You can modify numeric slice bounds in multiple ways:
 - Click the endpoints
 - Click the numbers above the number line
 - Drag the endpoints to readjust them

Create Custom Slices

Custom slices make it possible to group slices together into a single concept. This allows for custom logic to be configured within individual slices. Within these custom slices, you can decide what advanced logic (such as AND, OR, or NOT) connects the filters, similar to what can be done with population criteria. This feature is available for filter records where you can choose non-numeric values, such as diagnoses and encounter types.

1. Click in the **Slices** card.
2. Select the appropriate filter.
3. Click in the Slices card.
4. Click **Convert to Custom Slice**.

Grab only certain slices

For some measurements, grabbing the top or bottom values is not the most useful. For example, when investigating percentages, buckets with low numbers of records contributing to the numerator can skew the results. You can limit your slices to return only the buckets where another set of criteria are also met, like a count of patients of 50 or more. You can use any measure that you can create in SlicerDicer to restrict your slices.

To filter slices to a specific subset:

1. Add a slice.
2. Click next to the **Limit Your Slice Results** banner.
3. Click **Add Criterion** and select an additional measure.
4. Add lower and/or upper bounds to the measure.
5. Click **Update top slices?** to recalculate the slices.

ADD A MEASURE

Use measures to change how you're analyzing your population. By default, SlicerDicer shows the number of records that match the criteria you've specified. You can use measures to show rates or other metrics. Not all filters have the same measures available.

1. Click next to Measures.



- a. You can add as many measures as you like. Hide, show, or remove measures on your graph by hovering over the measure in the measures section and clicking the icons:  

2. Select a measure type.

Measure type	Definition
Percentage with selected value	The percentage of patients have the value you choose and match the other criteria. If a measure has multiple values documented during the query date range, the percentage calculation uses all values documented instead of only the most recent.
Sequential rate measure	Applies only to filters that track data over time. When you configure these filters as percentage measures, you can choose to make the percentage a sequential rate. This calculates the percentage of the filter value following or preceding another filter or slice value of your choice.
Percentage of sum	This measure divides the number of records in a slice by the number of records in the overall population. For example, suppose you use population criteria to show only records for patients with a legal sex of male, and then use slices to show the number of records in each department. The Percentage of Sum measure shows the number of male patients in each department divided by the total number of male patients.
Percentage of sliced sum	This measure applies only when there are two or more levels of slicing. It's similar to Percentage of Sum, but instead of dividing the values of each slice divided by the overall population, it divides the value of each subslice by the value of the slice the subslice is in. For example, suppose you slice by department and then slice by diagnosis. The Percentage of Sliced Sum measure shows the percentage of patients seen in each department with the specified diagnoses.



Maximum	The highest recorded value for a numeric filter.
Minimum	The lowest recorded value for a numeric filter.
Average	The mean of all recorded values for a numeric filter.
Deviation	The standard deviation of the recorded values for a numeric filter.
Variance	The deviation squared, which is also equal to the total difference between each patient's most recent value and the average value of a numeric filter.
Percentile	The value for which a specified percentage of records matching the other criteria is below.
Distinct count	This is the default measure for every data model. As the default measure, only distinct records are returned. Distinct count can also be configured on some criteria – in that case, it returns the number of unique values for that filter recorded in the date range. For example, a distinct count of Department would count the unique number of departments recorded on associated records in the session.
Risk ratio	The relative change in risk of an outcome event when comparing an exposure group to a reference group. The risk of an event is calculated as the total number experiencing the event divided by the total number in the group.
Odds ratio	The relative change in odds of an outcome event when comparing an exposure group to a reference group. The odds of an event is calculated as the total number experiencing the



	event divided by the total number not experiencing the event.
Custom formula	Formulas can be used to combine multiple measures into an overall calculation. Measures should be referenced by their tags and can be used with the following operators: +, -, *, /, and ().

3. Hover over results on the graph to view values for any measures you've added.

Add Goals and Thresholds to your measures

Thresholds and goals make it easier to home in on important data and parse large amounts of data at a glance. We recommend using thresholds or goals for SlicerDicer sessions with measures that have a specific target value or range. Thresholds allow users to set a color for any slices that fall within a range of values. Goals allow them to set a color for any slices that exceed a specific value and also appear as a horizontal dotted line on the graph.

To configure goals and/or thresholds for your measures:

- Click the measure you want to modify in the Measures section.
- Use the Add Goal or Add Threshold buttons to add your goals or thresholds.
- You can modify the specific values the goals or thresholds use to make them more meaningful.



GOALS AND THRESHOLDS DON'T APPEAR IN ALL GRAPH TYPES. THEY ALSO APPEAR ONLY WHEN THE GRAPH SHOWS A SINGLE MEASURE.

By default, goals appear as lines on your graph. You can hover over a goal line on the graph to view the definition. On the line graph, you can also choose to display the goals as a colored region instead of a line by modifying the Goal Style:

Goal Style



Line



Region

Since some values might be better when low, you can choose to display goal colors above the line using the **Lower is Better** checkbox on the measure.

Analyze the variability of your data with confidence intervals

You can add confidence intervals to average measures to quickly understand the variability of the data underlying the measure's value. This is useful since the amount and distribution of values can vary across different subsets of a population. For example, an average age at first diagnosis of 14 years could represent a group of teenagers or patients of all ages and might be based on 500 patients or 100,000 patients. SlicerDicer uses the Normal Approximation Method to calculate binomial confidence intervals of percentage measures. Statisticians sometimes refer to these as Wald confidence intervals.

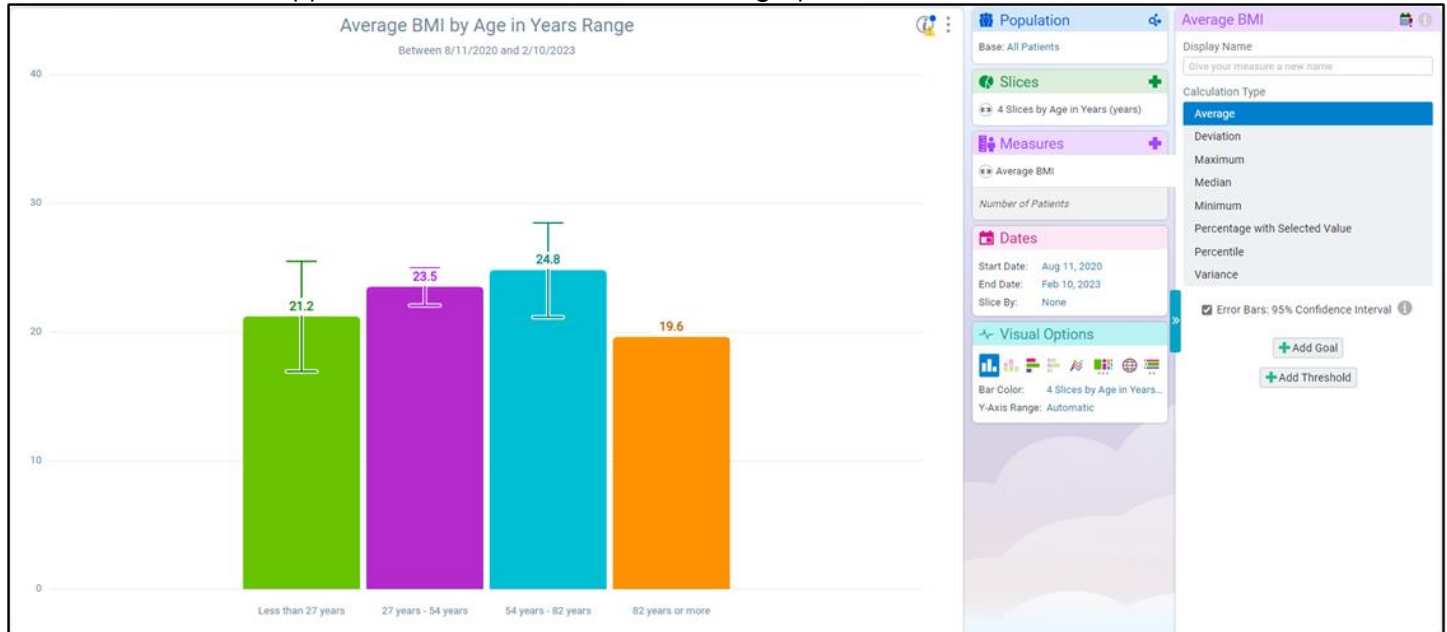
To add a confidence interval:

- Click an **Average** measure in the Measures section. Only Averages are supported currently.

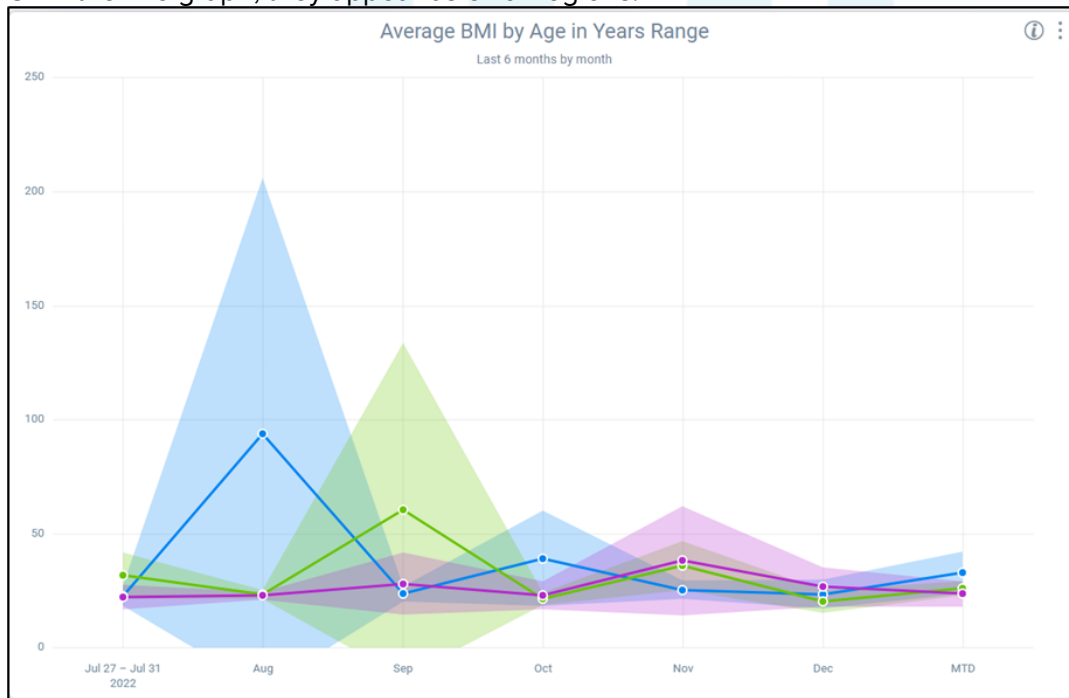


- Select the **Add Error Bars: 95% Confidence Interval** checkbox.

Confidence intervals appear on vertical and horizontal bar graphs as error bars:

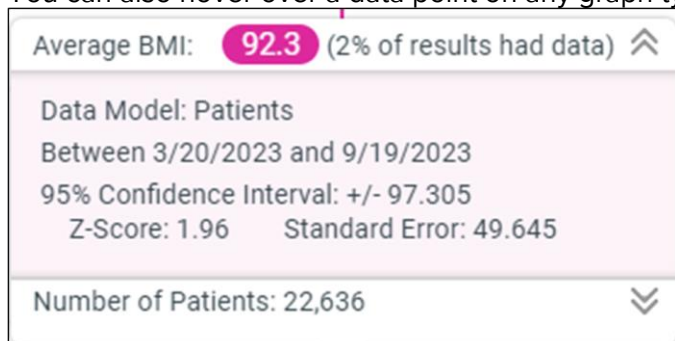


Or in the line graph, they appear as error regions:





You can also hover over a data point on any graph type to view the confidence interval and standard error.



Create Custom Measures

Custom measures make it possible to group multiple measures together into a single concept. This allows for custom logic to be configured within individual measures. Within these custom measures, you can decide what advanced logic (such as AND, OR, or NOT) connects the filters, similar to what can be done with population criteria. This feature is available for filter records where you can choose non-numeric values, such as diagnoses and encounter types.



CUSTOM MEASURES ARE ONLY AVAILABLE FOR PERCENTAGE OF SUM MEASURES.

1. Click  in the **Measures** card.
2. Select the **Percentage with Selected Value** measure type.
3. Choose your desired filter.
4. Click **Browse** in the measure card to search for a criterion to add to the measure.
5. Select your desired criterion. You now have a session with a custom measure.

CHANGE VISUALIZATION OPTIONS

Visual Options  control how your data is represented. Change elements like the graph type, measures, or colors to aid your analysis of the data.

Pick the right visualization for your analysis

You can change the graph type to better suit your data visualization. For example, click  to view a line graph. Available graphs are listed under Visual Options.

If you're not sure which graph type is best for your data, use this section as a quick reference:

- Vertical bar graph 
 - Compare different values and see differences at a glance.
 - Bar values are labeled.
- Horizontal bar graph 
 - Shows the same information as the vertical bar graph.
 - Shows crowded axis labels more clearly.



- 
 - Stacked bar graphs
 - See multiple slices together to group related concepts.
 - Hover over a bar to view specific values.
 - See both the sum of all slices and their individual values.
- 
 - Line graph
 - See how results change over time. This visualization is not available if the session date range is All Time.
 - You can change the date summary level by clicking the Dates section.
 - You can modify how the line graph displays unavailable data, add line labels, and modify how goals are displayed in the line graph in the Visual Options section.
- 
 - Tree map
 - Group hierarchical data together.
 - Use the whole area of the graph to show shapes more clearly than bar graphs, such as when you have multiple levels of slices.
 - Note that it's harder to compare area than it is to compare height and length, so it can be more difficult to spot differences between two values than when using a bar graph.
- 
 - Map visualization
 - See if geographic area is correlated with your results.
 - Note that maps aren't available for every data model or at every organization.
- 
 - Detail table
 - View summarized details about your results.
 - View line-level details for records to which you have access.
 - Export results to Excel for further analysis. You can choose to export line-level data (all data in the table) or summary data (only slice and date group data).
 - In some data models, you can launch an activity, like opening a patient chart or surgery preference card.
 - You can add, remove, or rearrange columns by hovering, right-clicking, or dragging the column header bar.
- 
 - Cross Tab
 - View relationships between slices by grouping them across columns and rows.
 - Similar to a pivot table or heatmap.
 - Supports multi-column sorting, collapsing and expanding slices, and drilling down to line-level details.

View details about records in the Detail Table

In certain data models, you can view additional real-time information about detail table results. You can see additional information about related records without needing to leave SlicerDicer or export the information to another application. Not all data models support this feature. Data models with this extra detail include AP Claims, Lab Specimens and Tests, Births, Patient Infections, Anesthesia Records, and more.



To view details about a record:

1. In your SlicerDicer session, switch to the detail table and expand some rows.
2. Select a row in the detail table.
3. At the bottom of the screen, click .
4. A report opens, showing you additional information about the selected row.

See relationships with the Crosstab visualization

The crosstab visualization allows you to see relationships between slices by grouping slices across columns and rows. You can see the "cross" between these data points.

1. Click the detail table in the **Visual Options** section.
2. Select the Crosstab **Table Type**.
 - a. By default, the first slice is assigned as the column slice, with the second slice, and any subsequent slices, assigned as row slices.
3. Click **Add row** or **Add column** to add additional slices to the Crosstab table.
 - a. To better interpret results, you can use a measure to change the default cell colors. Under **Cell Color**, click **None** and select an existing measure.
4. To sort, click a column header.

See populations on a map

Use the map visualization to investigate whether data correlates geographically, such as by census tract, postal code, county, state, or country. Darker areas on the map have a higher amount of whatever metric you're measuring.

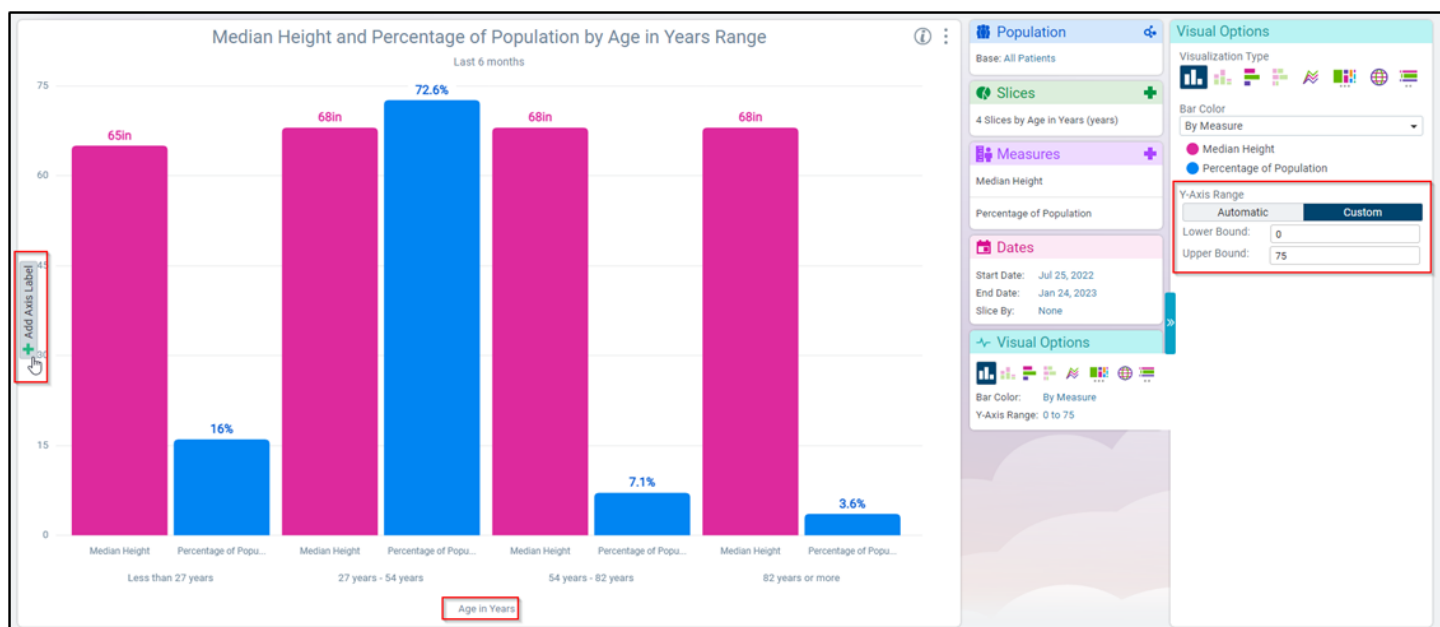
Most data models in SlicerDicer use the patient's current permanent address as the location source. For example, in the ED Encounters data model, the colors on the map correspond to where patients live, not to where the ED visits occurred. However, some data models have additional location sources, like Encounter Department.

1. Click the globe  to open the map.
2. Hover over a geographic boundary to see more details about the population, like the raw count or percentage of patients in that boundary who meet the criteria you specified.
3. To analyze data on the map:
 - Pan around the map or zoom in or out to see if there are similar patterns in other areas.
 - Click the magnifying glass  and then enter a ZIP Code to center the map on that ZIP Code.
 - Be aware that the map can only show one slice and one measure at a time.

Modify the Graph's Axis Definitions

SlicerDicer automatically calculates the Y axis of the graph based on the values of the measures returned in your session. Sometimes, you might want to use a different axis scale to view your data. Or you might want a label to describe what each axis represents. You might want to do this if you have multiple measures with large differences in scale. It is also useful if you share the session with other users.

1. To change your Y axis scale, click the **Visual Options** section or click the link after the Y-Axis Range header.
 - a. Switch the axis from Automatic to Custom.
 - b. Add new custom upper and lower bounds.
2. To add axis labels:
 - a. Hover over the graph near the bottom center (X axis) or left center (Y axis).
 - b. Click the Add Axis Label button and enter your description.



Customize your graph colors

You can customize the colors for slices and measures. You can also choose what the graph is using as a color. Any given slice or measure in your session is available to color the graph, even when hidden.

To customize individual colors for slices, click a colored circle. You can choose from a preset list of standard colors or choose a custom color using the Advanced Configuration tool.

Semantic

Neutral

Colorful

Advanced Configuration

Slice by Department

5

Grab Top 5

Grab Bottom 5

Limit Your Slice Results

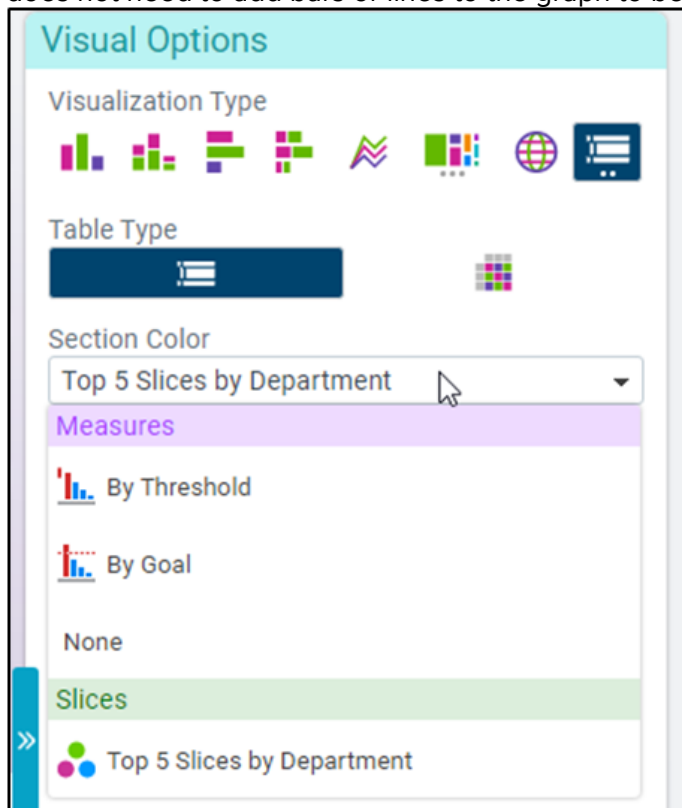
Current Slices

- ☒ OB L&D
- ☐ OB NURSERY
- ☐ WI HBN INTERNAL MED
- ☐ LAKEVIEW MED SURG
- ☐ INITIAL DEPARTMENT
- ☐ None of the above

You can color your graph by slice, measure, goal, or threshold, depending on what you have configured in your session. To change the current graph color, click the Visual Options header or click the Color link. If you have



multiple measures, coloring by one measure allows you to save space on the graph, since the Color By measure does not need to add bars or lines to the graph to be displayed.



HIDE NONE OF THE ABOVE, SAVE CLICKS, AND SHARE DATA

SlicerDicer allows you to customize certain parts of your experience. To find the available settings for your user, click  **Settings** in the toolbar.

- You can control whether a "None of the above" slice appears automatically when you add slices to a session.
- You can enable or disable certain reminders when you add certain types of content, like Last Stored filters, None of the Above slices, and more.



- You can also choose to share your data with other users. Note that this could expose sensitive information, so you should not make this decision lightly.

SLICERDICER SHORTCUTS

SlicerDicer has many shortcuts that are different from those you might be used to in other parts of Hyperspace.

You can view a list of supported date shortcuts by clicking the calendar icon in the Start Date or End Date fields and choosing **Shortcuts**.

Always Applicable

Press:

- Ctrl + Z** to undo your last action.
- Ctrl + Y** to redo the last action you undid.

Use these shortcuts to select multiple items from a list:

- Ctrl + click** to select multiple items one at a time. Select an item again while pressing **Ctrl** to clear it.
- Shift + click** the first and last items in a list to select them along with all the items in between.

Text Editing

Press:

- Ctrl + A** to select all text in a text box.



- **Ctrl + C** to copy selected text.
- **Ctrl + V** to paste the text you copied.
- **Ctrl + X** to cut selected text.

User Interface

Press:

- **Esc** to close a popup window.
- **Ctrl + Up Arrow** or **Ctrl + Down Arrow** to reorder slices in the slice editor.
- **Up Arrow** or **Down Arrow** to select different measure types in the measure editor when you're in the measure type list under the “Calculation Type” header. These shortcuts are available only if the filter supports multiple measure types.
- **Left Arrow** or **Right Arrow** to select different visualization types in the Visual Options editor when you're on one of the visualization icons.
- **Shift + Enter** to open or close the Report Viewer (Available starting in May 2022).
- **≡** or **Right-click** on a criterion to open the Advanced Options menu.
- **≡** or **Right-click** on the graph to open a menu of common actions.

Detail Table Shortcuts

The detail table has shortcuts not found elsewhere in SlicerDicer. To use these shortcuts, choose the Detail Table visualization and move your focus to a row in the table.

Press:

- **Home** to move to the top row in the table.
- **End** to move to the bottom row in the table.
- **Up Arrow** or **Down Arrow** to select a specific row in the table.
- **Page Up** or **Page Down** to move through the table quickly.
- **Alt + Up Arrow** or **Alt + Down Arrow** to move between summary rows in the table.
- **Shift + Up Arrow** or **Shift + Down Arrow** to select a range of rows in the table.
- **Shift + Page Up** or **Shift + Page Down** to select a large range of rows in the table quickly.
- **Ctrl + Space** to select a row or clear your selected row.



- **Ctrl + A** to select all rows in the table.
- **Ctrl + Up Arrow** or **Ctrl + Down Arrow** to avoid selecting rows while moving through the table.
- **Ctrl + Page Up** or **Ctrl + Page Down** to avoid selecting rows while moving through the table quickly.

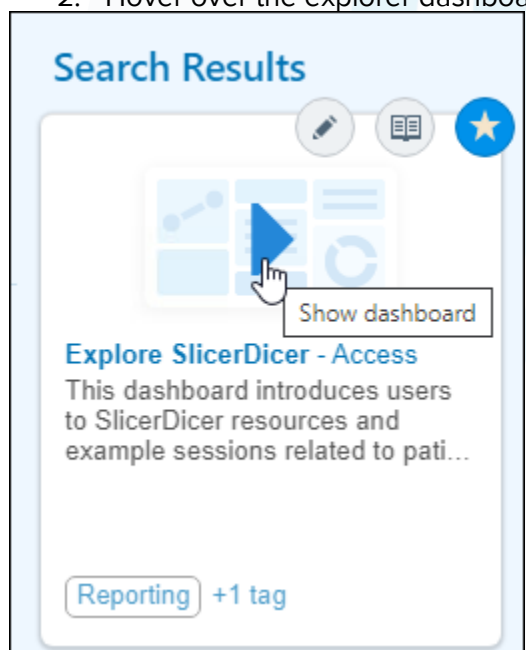
DISCOVER SLICERDICER USING EXPLORER DASHBOARDS

Explorer dashboards help you get familiar with how to use SlicerDicer. Each dashboard contains links to additional documentation as well as several example sessions that demonstrate what kinds of reports you can create with SlicerDicer.

Try It Out

This tip sheet walks you through using an Explore SlicerDicer dashboard. There are explorer dashboards for clinical, patient access, and HIM use cases. Which explorer dashboards are available to you depends on whether you have access to reporting content for those areas. Keep in mind that the data at your organization won't match the data in these examples.

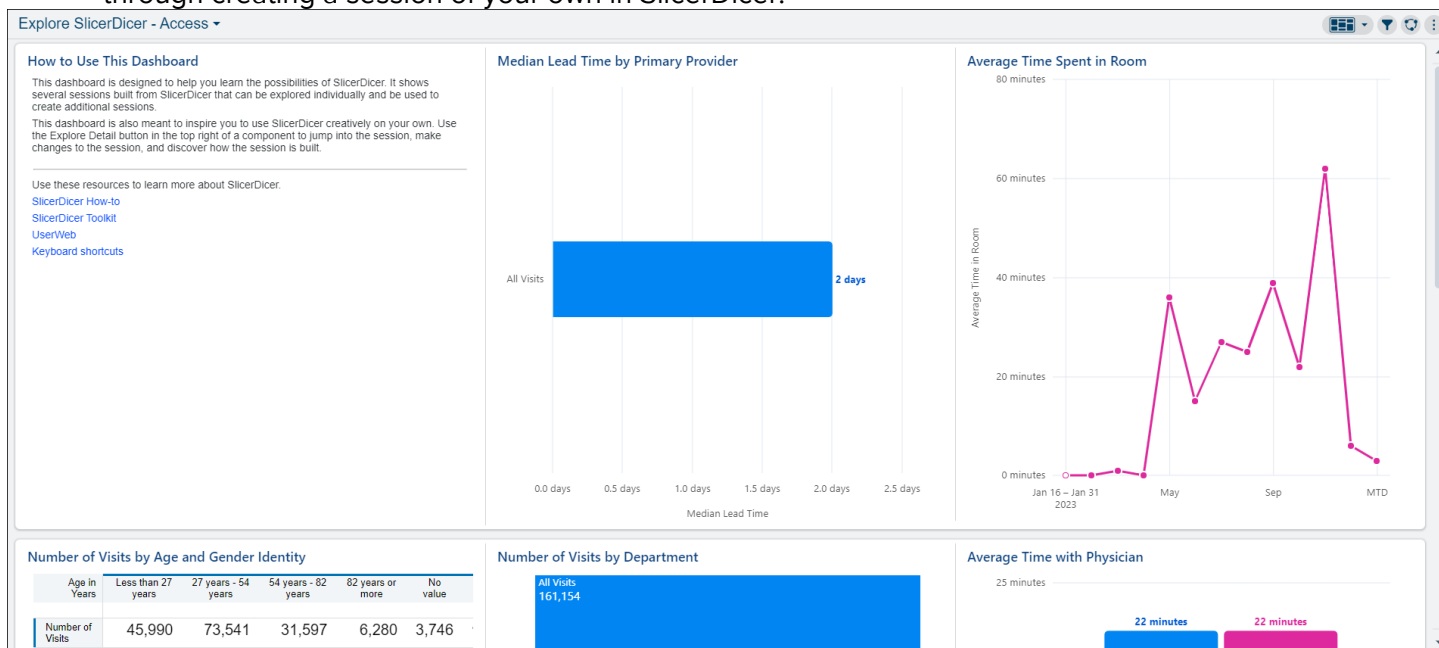
1. In Hyperspace, go the Analytics Catalog and search for Explore SlicerDicer to see which explorer dashboards are available to you.
2. Hover over the explorer dashboard you want to view and click **Show dashboard**.



3. The explorer dashboard loads. Instructions for how to use the dashboard appear at the top left, along with a list of additional resources you might find useful.
4. Next, review the example reports created for popular uses cases. For example, the patient access explorer dashboard includes reports for the median lead time by primary provider at your organization, number of visits by department, and average time with physician.



5. Hover over a session you're interested in and click  to open the session in SlicerDicer.
6. When you're done exploring the session, click **Cancel** to return to the explorer dashboard.
7. Scroll to the very bottom of the dashboard to find the **How to Build It** section. This section walks you through creating a session of your own in SlicerDicer.



Additional Resources

You might also be interested in this additional resource:

- SlicerDicer tutorial. When you open SlicerDicer for the first time, you have the option to complete an in-system tutorial. If you close the tutorial, you can return to it at any time by clicking **Tutorials** in the top-right corner.



Date Modified and Version	What Changes	Modified by:	Reviewed By:
V1 6/4/2024	Converted from Epic	Amy Gifford	

